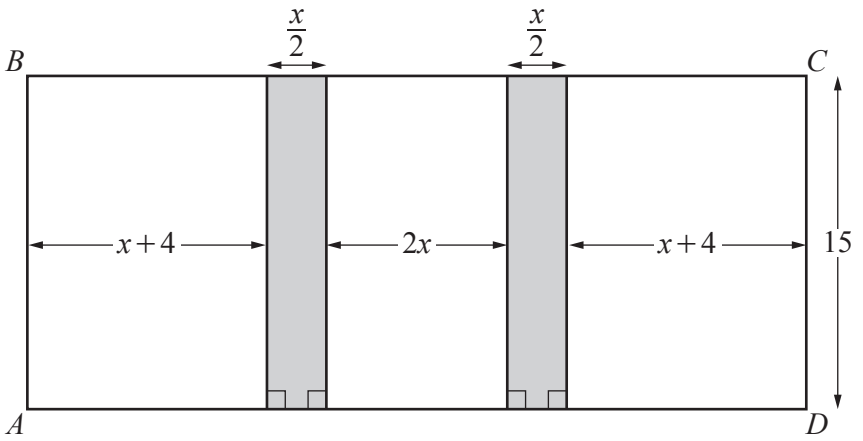
0580_w22_qp_33_Q5

5 **(a)** In this part, all measurements are in centimetres.



NOT TO SCALE

The diagram shows a rectangle, *ABCD*.

(i) Show that the length *AD* is $5x+8$.

[1]

(ii) The area of *ABCD* is $360\,\text{cm}^2$.

Work out the value of x.

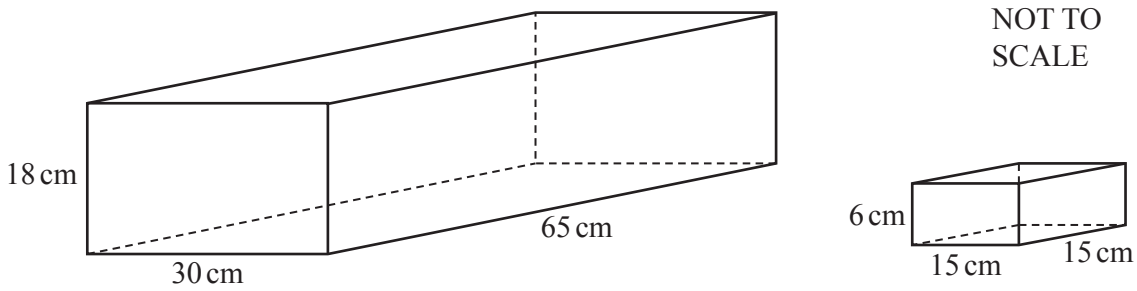
$x =$ ..... [4]

(iii) Find the total shaded area.

..... cm^2 [1]

0580_w22_qp_33_Q5

(b)



Rectangular box

Cuboid

The diagram shows an open rectangular box and a solid cuboid.

(i) Show that a maximum of 24 of these cuboids will fit inside the box.

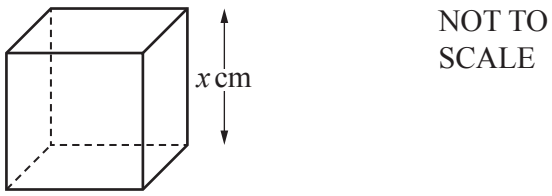
[1]

(ii) 24 of these cuboids are placed inside the box.

Calculate the volume of empty space in the box.
Give the units of your answer.

.... [4]

(c)



The diagram shows a solid cube with side x cm.
The total surface area of the cube is $486\,\text{cm}^2$.

Calculate the value of x.

$x =$ ..... [2]